

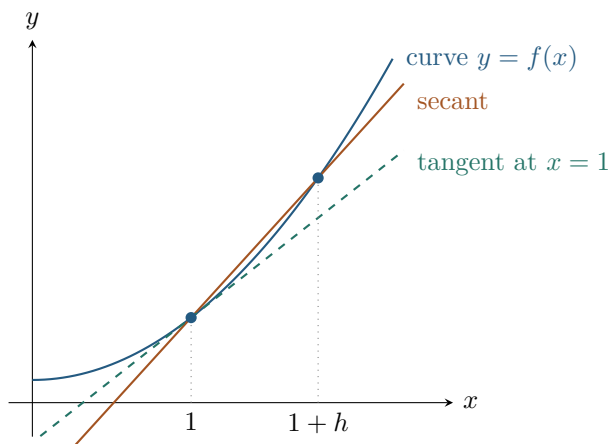
## Limits

MATH 1334

September 28, 2026

**Goal.** Compute slopes of tangent lines.

A *secant line* passes through two points on the graph. Its slope is the average rate of change. A *tangent line* gives the instantaneous rate of change at a point (instantaneous velocity for a position function).



Secant slope:

$$\frac{f(1+h) - f(1)}{h}, \quad h \neq 0.$$

More generally, the *difference quotient* is

$$\frac{f(a+h) - f(a)}{h}.$$

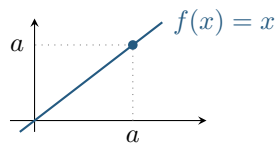
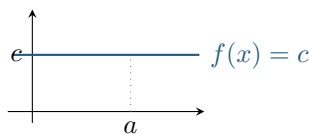
As  $h \rightarrow 0$ , the second point approaches the first. We study limits to describe what the secant slope approaches.

**Definition.** We say  $f$  is *defined near*  $a$  if it is defined on some open interval containing  $a$ , except possibly at  $a$ . For example,  $(x^2 - 1)/(x - 1)$  is defined near 1, although it is undefined at 1.

**Definition (informal).** We write  $\lim_{x \rightarrow a} f(x) = L$  if the values of  $f(x)$  can be made arbitrarily close to  $L$  by taking *all*  $x$  sufficiently close to  $a$ , with  $x \neq a$ . We also write  $f(x) \rightarrow L$  as  $x \rightarrow a$ . If no such real number  $L$  exists, the finite limit does not exist (DNE).

**Theorem.** For real constants  $a$  and  $c$ ,

$$\lim_{x \rightarrow a} c = c, \quad \lim_{x \rightarrow a} x = a.$$



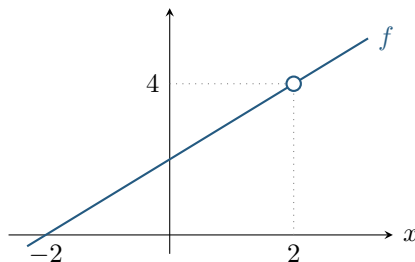
## Limits and function values

*Example 1.* Let  $f(x) = \frac{x^2 - 4}{x - 2}$  and  $a = 2$ .

*Graph method.* For  $x \neq 2$ , factoring gives

$$f(x) = \frac{(x-2)(x+2)}{x-2} = x+2.$$

The graph is the line  $y = x + 2$  with a hole at  $(2, 4)$ . From either side of 2, the graph approaches height 4.



*Table method.* Choose inputs approaching 2 from both sides; do not substitute  $x = 2$  into the original fraction.

|         |     |      |       |         |     |      |       |
|---------|-----|------|-------|---------|-----|------|-------|
| $x < 2$ | 1.9 | 1.99 | 1.999 | $x > 2$ | 2.1 | 2.01 | 2.001 |
| $f(x)$  | 3.9 | 3.99 | 3.999 | $f(x)$  | 4.1 | 4.01 | 4.001 |

The table suggests the same limit. A table is evidence, not a proof; the identity  $f(x) = x + 2$  for  $x \neq 2$  justifies the conclusion:

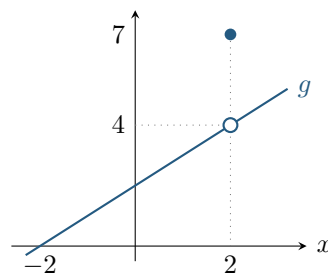
$$\lim_{x \rightarrow 2} f(x) = 4, \quad f(2) \text{ is undefined.}$$

*Example 2.* Let  $g(x) = \begin{cases} x + 2, & x \neq 2, \\ 7, & x = 2. \end{cases}$

*Graph method.* The nearby graph is still the line  $y = x + 2$ . The filled point at  $(2, 7)$  changes only the value at 2:

$$\lim_{x \rightarrow 2} g(x) = 4, \quad g(2) = 7 \neq 4.$$

A limit describes nearby values, not necessarily the value at the point.



## One-sided limits

**Definition.** The *left-hand limit*  $\lim_{x \rightarrow a^-} f(x) = L$  means  $f(x) \rightarrow L$  as  $x$  approaches  $a$  through values  $x < a$ . The *right-hand limit*  $\lim_{x \rightarrow a^+} f(x) = L$  uses values  $x > a$ .

**Theorem.** The two-sided limit exists and equals  $L$  exactly when both one-sided limits equal  $L$ :

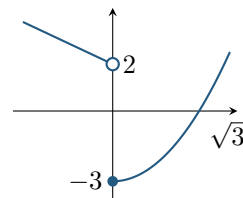
$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L.$$

*Example 1.*  $f(x) = \begin{cases} 2 - x, & x < 0, \\ x^2 - 3, & x \geq 0. \end{cases}$

*Case method.* Use  $2 - x$  on the left and  $x^2 - 3$  on the right:

$$\lim_{x \rightarrow 0^-} f(x) = 2, \quad \lim_{x \rightarrow 0^+} f(x) = -3.$$

*Graph method.* The branches approach different heights. The one-sided limits differ, so  $\lim_{x \rightarrow 0} f(x)$  DNE.

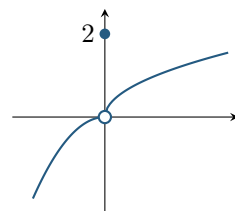


*Example 2.*  $g(x) = \begin{cases} -x^2, & x < 0, \\ 2, & x = 0, \\ \sqrt{x}, & x > 0. \end{cases}$

*Case method.* Use  $-x^2$  on the left and  $\sqrt{x}$  on the right:

$$\lim_{x \rightarrow 0^-} g(x) = \lim_{x \rightarrow 0^+} g(x) = 0.$$

*Graph method.* Both branches approach height 0. Thus  $\lim_{x \rightarrow 0} g(x) = 0$ , although  $g(0) = 2$ .



*Example 3.* Find  $\lim_{x \rightarrow 2} h(x)$  for  $h(x) = \frac{x^2 - 4}{|x - 2|}$ .

*Table method.* This second table illustrates why checking both sides matters.

|         |      |       |        |         |     |      |       |
|---------|------|-------|--------|---------|-----|------|-------|
| $x < 2$ | 1.9  | 1.99  | 1.999  | $x > 2$ | 2.1 | 2.01 | 2.001 |
| $h(x)$  | -3.9 | -3.99 | -3.999 | $h(x)$  | 4.1 | 4.01 | 4.001 |

*Case method.* Since  $|x - 2| = -(x - 2)$  for  $x < 2$  and  $|x - 2| = x - 2$  for  $x > 2$ ,

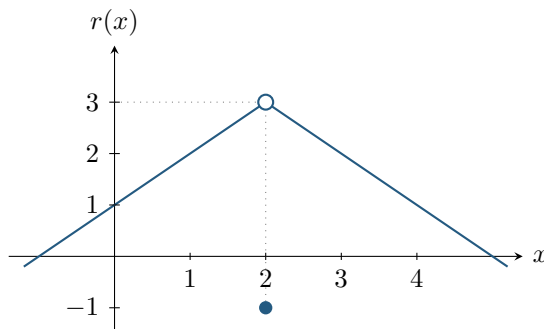
$$h(x) = \begin{cases} -(x + 2), & x < 2, \\ x + 2, & x > 2, \end{cases} \quad \lim_{x \rightarrow 2^-} h(x) = -4, \quad \lim_{x \rightarrow 2^+} h(x) = 4.$$

Therefore  $\lim_{x \rightarrow 2} h(x)$  DNE.

## Practice

1. Compute  $r(2)$ ,  $\lim_{x \rightarrow 2^-} r(x)$ ,  $\lim_{x \rightarrow 2^+} r(x)$ , and  $\lim_{x \rightarrow 2} r(x)$ , where

$$r(x) = \begin{cases} x + 1, & x < 2, \\ -1, & x = 2, \\ 5 - x, & x > 2. \end{cases}$$



*Graph method.* The filled point gives  $r(2) = -1$ . Both branches approach the open point at height 3, so

$$\lim_{x \rightarrow 2^-} r(x) = 3, \quad \lim_{x \rightarrow 2^+} r(x) = 3, \quad \lim_{x \rightarrow 2} r(x) = 3.$$

2. Compute  $\lim_{x \rightarrow 3^-} s(x)$ ,  $\lim_{x \rightarrow 3^+} s(x)$ , and  $\lim_{x \rightarrow 3} s(x)$  for

$$s(x) = \frac{x^2 - 9}{|x - 3|}.$$

*Case method.* Absolute values require cases:

$$\begin{aligned} x < 3 : \quad s(x) &= \frac{(x - 3)(x + 3)}{-(x - 3)} = -(x + 3), \\ x > 3 : \quad s(x) &= \frac{(x - 3)(x + 3)}{x - 3} = x + 3. \end{aligned}$$

Thus

$$\lim_{x \rightarrow 3^-} s(x) = -6, \quad \lim_{x \rightarrow 3^+} s(x) = 6,$$

and the two-sided limit DNE.

3. **True or false.** If  $f(a)$  is undefined, then  $\lim_{x \rightarrow a} f(x)$  DNE.

*Solution.* False. In Example 1 on page 2,

$$f(x) = \frac{x^2 - 4}{x - 2}$$

is undefined at 2, but  $\lim_{x \rightarrow 2} f(x) = 4$ .

## Infinite limits and vertical asymptotes

**Definition.** We write  $\lim_{x \rightarrow a} f(x) = \infty$  if outputs exceed every positive bound for all sufficiently nearby  $x \neq a$ .

We write  $\lim_{x \rightarrow a} f(x) = -\infty$  if outputs fall below every negative bound for all sufficiently nearby  $x \neq a$ .

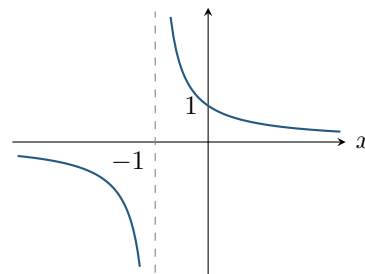
Similarly for  $x \rightarrow a^-$  and  $x \rightarrow a^+$ . Infinity describes unbounded behavior; it is not a real-number limit.

The line  $x = a$  is a *vertical asymptote* if at least one of the one-sided limits at  $a$  is  $\infty$  or  $-\infty$ .

*Example 1.*  $A(x) = \frac{1}{x+1}, \quad a = -1.$

$$\lim_{x \rightarrow -1^-} A(x) = -\infty, \quad \lim_{x \rightarrow -1^+} A(x) = \infty.$$

The two-sided limit DNE. The line  $x = -1$  is a vertical asymptote.



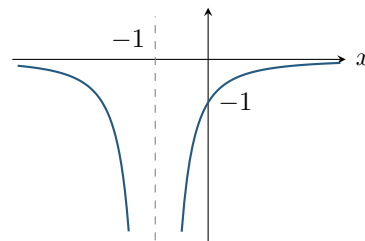
*Example 2.*  $B(x) = \frac{-1}{(x+1)^2}.$

$$\lim_{x \rightarrow -1^-} B(x) = -\infty, \quad \lim_{x \rightarrow -1^+} B(x) = -\infty.$$

Both sides have the same unbounded behavior:

$$\lim_{x \rightarrow -1} B(x) = -\infty.$$

Again,  $x = -1$  is a vertical asymptote.



**Theorem.** For a positive even integer  $n$ ,

$$\lim_{x \rightarrow a} \frac{1}{(x-a)^n} = \infty.$$

For a positive odd integer  $n$ ,

$$\lim_{x \rightarrow a^-} \frac{1}{(x-a)^n} = -\infty, \quad \lim_{x \rightarrow a^+} \frac{1}{(x-a)^n} = \infty.$$

## Harder limits

1. Find  $\lim_{x \rightarrow 0} \frac{x-1}{x^2(x+1)}$ .

*Case method.* For  $x$  sufficiently close to 0,  $x-1 < 0$ ,  $x+1 > 0$ , and  $x^2 > 0$  when  $x \neq 0$ . The numerator approaches  $-1$  and the denominator approaches 0 through positive values on both sides. Hence

$$\lim_{x \rightarrow 0^-} \frac{x-1}{x^2(x+1)} = -\infty, \quad \lim_{x \rightarrow 0^+} \frac{x-1}{x^2(x+1)} = -\infty,$$

so  $\lim_{x \rightarrow 0} \frac{x-1}{x^2(x+1)} = -\infty$ .

2. Find  $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ , with  $x$  in radians.

*Table method.* Numerical values help suggest the answer here.

| $x$        | $\pm 0.1$ | $\pm 0.01$ | $\pm 0.001$ |
|------------|-----------|------------|-------------|
| $\sin x/x$ | 0.998334  | 0.999983   | 1.000000    |

The entries are rounded to six decimal places. Taking  $x_n = 1/n$  and  $x_n = -1/n$  suggests the same value from both sides:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

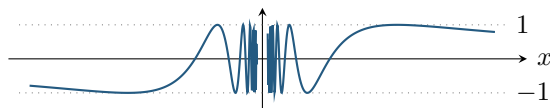
This table suggests the result; a proof can be given using the squeeze theorem.

3. Find  $\lim_{x \rightarrow 0} \sin(1/x)$ .

*Graph method.* The function keeps oscillating between  $-1$  and  $1$ . For positive integers  $n$ , let

$$x_n = \frac{1}{\pi/2 + 2\pi n}, \quad y_n = \frac{1}{3\pi/2 + 2\pi n}.$$

Both approach 0, but  $\sin(1/x_n) = 1$  and  $\sin(1/y_n) = -1$ . Therefore the limit DNE.

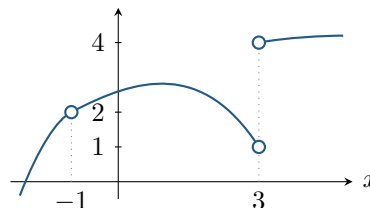


## Limits to graphs

Construct a graph with the following properties:

$$\begin{aligned} \lim_{x \rightarrow -1} u(x) &= 2, & \lim_{x \rightarrow 3^-} u(x) &= 1, \\ \lim_{x \rightarrow 3^+} u(x) &= 4, & u(3) &\text{undefined.} \end{aligned}$$

One possible graph is shown. A hole at  $x = -1$  is allowed but not required.



## Practice

1. Let  $F(x) = \frac{-2}{(x-1)^3}$ . Find  $\lim_{x \rightarrow 1} F(x)$ .

*Case method.* Since the power is odd,

$$\lim_{x \rightarrow 1^-} \frac{1}{(x-1)^3} = -\infty, \quad \lim_{x \rightarrow 1^+} \frac{1}{(x-1)^3} = \infty.$$

Multiplication by  $-2$  reverses the signs:

$$\lim_{x \rightarrow 1^-} F(x) = \infty, \quad \lim_{x \rightarrow 1^+} F(x) = -\infty.$$

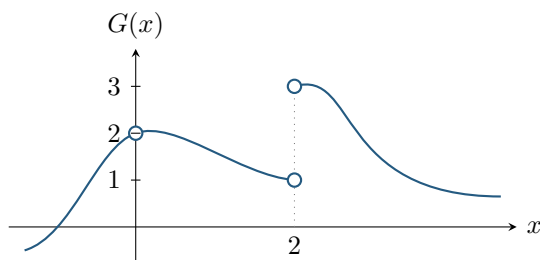
Therefore  $\lim_{x \rightarrow 1} F(x)$  DNE.

2. Construct a graph satisfying

$$\lim_{x \rightarrow 0} G(x) = 2, \quad \lim_{x \rightarrow 2^-} G(x) = 1, \quad \lim_{x \rightarrow 2^+} G(x) = 3,$$

with  $G(2)$  undefined.

*Solution.* One possible graph is shown below. Its left branch approaches height 1 at  $x = 2$ ; its right branch approaches height 3. Neither endpoint at  $x = 2$  is included.

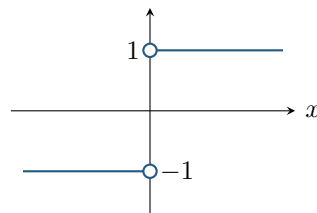


The hole at  $(0, 2)$  is optional: the condition at 0 specifies a limit, not a function value.

3. **True or false.** If  $\lim_{x \rightarrow a} f(x)$  DNE, then the function approaches  $\infty$  or  $-\infty$ .

*Solution.* False. For example,

$$f(x) = \begin{cases} 1, & x > 0, \\ -1, & x < 0. \end{cases}$$



At  $a = 0$ , the left-hand limit is  $-1$  and the right-hand limit is  $1$ . The two-sided limit DNE, yet the function remains bounded.